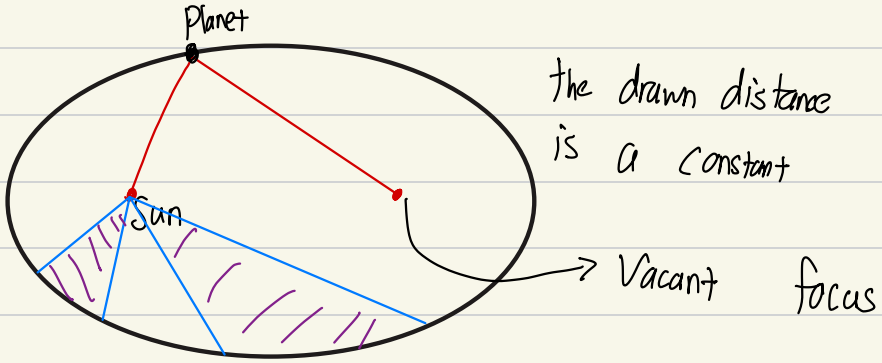
9/18 Kepler's Laws

Johannes Kepler: 1571 - 1630

1st law (1609): The orbit of a planet is an ellipse, with the Sun located at one of its foci



2nd law (1609): The line joining the planet and the Sun sweeps out equal areas in equal amounts of time

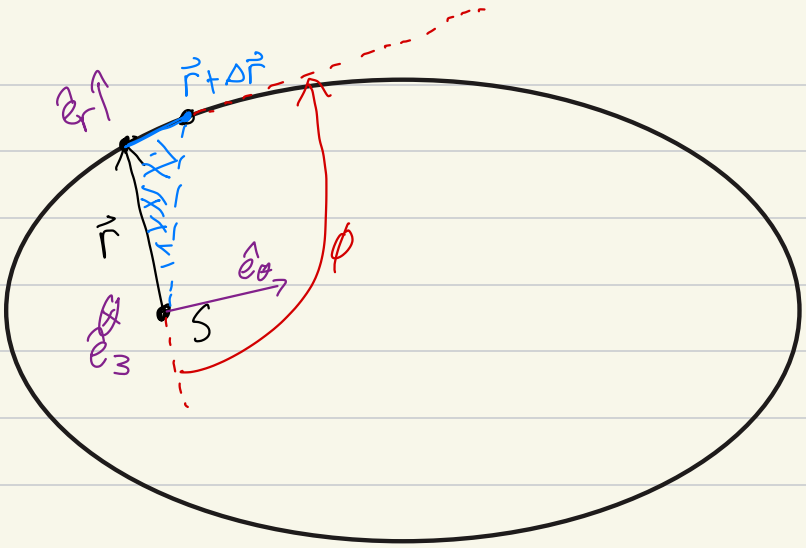
3rd law (1619): The square of a planet's orbital period (year) is proportional to the cube of the planet's mean distance from the sun

$$T^2 \sim R^3$$

T: length of year

R: average radius

Newton (1643-1727) will help us understand why these laws are valid - and explain even more



Swept area $= \frac{1}{2}|\vec{r}|\,|\Delta\vec{r}|\sin\phi$

(Areal) $\hat{e}_3 = \frac{1}{2}\vec{r}\times\Delta\vec{r}$ $\qquad \frac{1}{2}\vec{r}\times\Delta\vec{r} = \hat{e}_3\,\Delta\text{area} = \text{Const}$

divide by Δt

$$\frac{1}{2}\vec{r}\times\frac{\Delta\vec{r}}{\Delta t} = \hat{e}_3\frac{\Delta\text{area}}{\Delta t} \rightarrow \frac{1}{2}(\vec{r}\times\vec{v}) = \hat{e}_3\frac{d}{dt}\text{Area} \quad ; \quad \vec{r}\times\vec{v} = \text{Const}$$

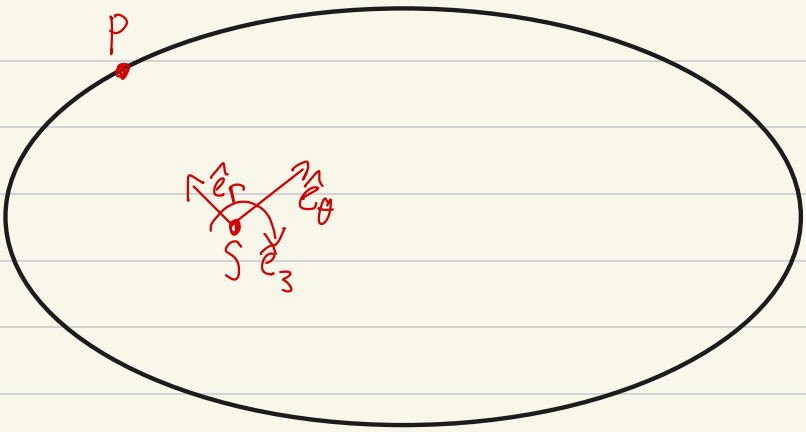
Define ref fram $E\{S, \hat{e}_r, \hat{e}_\theta, \hat{e}_3\}$ $\qquad \vec{r} = r\hat{e}_3 \quad ; \quad r = |\vec{r}|$

$$\vec{v} = \dot{\vec{r}} = \dot{r}\hat{e}_r + r\dot{\hat{e}}_r = \dot{r}\hat{e}_r + r(\vec{w}^{E/N}\times\hat{e}_r)$$

$$\vec{v} = \dot{r}\hat{e}_r + r\dot{\theta}\hat{e}_\theta$$

$$\vec{w}^{E/N} = \dot{\theta}\hat{e}_3$$

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$\vec{r} = r\hat{e}_r$

$\vec{v} = \dot{r}\hat{e}_r + r\dot{\theta}\hat{e}_\theta$

$\vec{r} \times \vec{v}$ conserved, $\vec{r}\times\vec{v} = r\hat{e}_r \times (\dot{r}\hat{e}_r + r\dot{\theta}\hat{e}_\theta) = r^2\dot{\theta}\hat{e}_3$ ($\dot{r}\hat{e}_r \times \hat{e}_r = 0$)

$\vec{h} = \vec{r}\times\vec{v}$ is, Specific angular momentum ; Specific = per unit mass

Angular Momentum $\vec{H} = \vec{r}\times(m\vec{v})$, $\vec{h} = \frac{\vec{H}}{m} = \vec{r}\times\vec{v}$

What causes $\vec{H}$ to change

$\frac{d\vec{H}}{dt} = \vec{v}\times m\vec{v} + \vec{r}\times m\vec{a}$ ($\vec{v}\times m\vec{v} = 0$)

$\frac{d\vec{H}}{dt} = \vec{r}\times\vec{F} = 0$ for an orbit

$r\hat{e}_r \times \vec{F} = 0$

F is parallel to $\hat{e}_r$

force acting on the planet are radial

$$\vec{a} = \overset{U}{\frac{d}{dt}}\vec{v} = \overset{N}{\frac{d}{dt}}\left(\dot{r}\hat{e}_r + r\dot{\theta}\hat{e}_\theta\right) = \ddot{r}\hat{e}_r + \dot{r}\dot{\hat{e}}_r + \dot{r}\dot{\theta}\hat{e}_\theta + r\ddot{\theta}\hat{e}_\theta + r\dot{\theta}\dot{\hat{e}}_\theta \rightarrow$$

$$\dot{\hat{e}}_r = \vec{w}^{E/N} \times \hat{e}_r = \dot{\theta}\hat{e}_3 \times \hat{e}_r = \dot{\theta}\hat{e}_\theta$$

$$\dot{\hat{e}}_\theta = \vec{w}^{E/N} \times \hat{e}_\theta = \dot{\theta}\hat{e}_3 \times \hat{e}_\theta = -\dot{\theta}\hat{e}_r$$

$$\vec{a} = (\ddot{r} - r\dot{\theta}^2)\hat{e}_r + (\cancel{2\dot{r}\dot{\theta} + r\ddot{\theta}}^{=0})\hat{e}_\theta$$

$\frac{d}{dt}(r^2\dot{\theta}) = 0$ → we know

$$\underbrace{(2\dot{r}\dot{\theta} + r\ddot{\theta})}_{0} r = 0$$

acceleration in radial direction ($\hat{e}_r$)

$\vec{F} = F\hat{e}_r$, $f = \frac{F}{m}$ $\qquad f = \frac{F}{m} = \ddot{r} - r\dot{\theta}^2$

We have $\vec{h} = \vec{r} \times \vec{v} = r^2\dot{\theta}\hat{e}_3$

$$h\hat{e}_3 = \vec{h}$$

$$h = r^2\dot{\theta}$$

$$\dot{\theta} = \frac{h}{r^2}$$

$f = \ddot{r} - \frac{h^2}{r^3}$ (eq 1) $\qquad$ Let $r = \frac{1}{u}$, $u = \frac{1}{r}$

Remark: $\frac{dy}{dt} = \frac{dy}{d\theta}\frac{d\theta}{dt}$

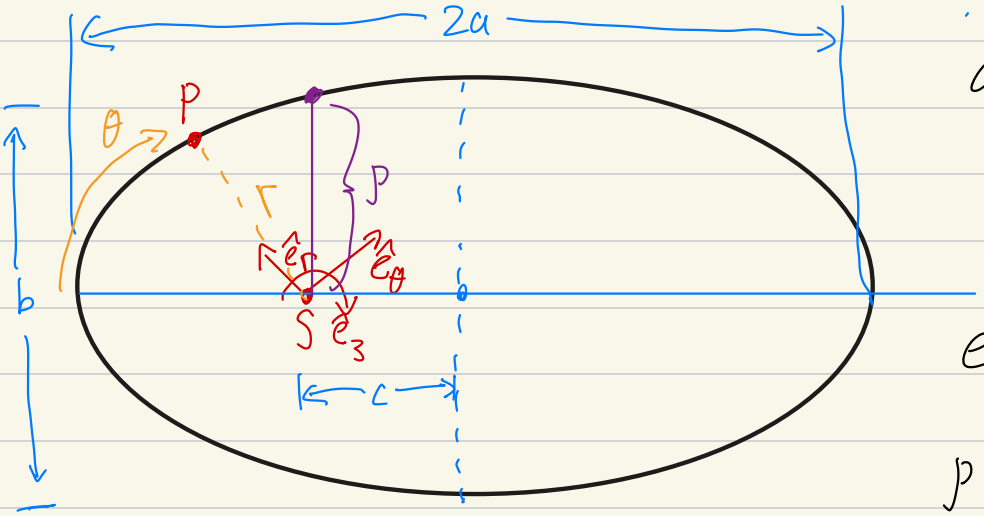
$$\frac{dr}{dt} = \frac{-1}{u^2}\frac{du}{dt} \cdot \frac{d\theta}{dt} = \underbrace{-r^2\dot{\theta}}_{-h}\frac{du}{d\theta} = -h\frac{du}{d\theta}$$

$$\frac{d^2r}{dt^2} = -h\frac{d}{dt}\left(\frac{du}{d\theta}\right) = -h\frac{d}{d\theta}\left(\frac{du}{d\theta}\right)\frac{d\theta}{dt} = -h\dot{\theta}\frac{d^2u}{d\theta^2}$$

$$f = -h\dot{\theta}\frac{d^2u}{d\theta^2} - h^2u^3 \qquad h = r^2\dot{\theta} = \text{const}$$

$$h = \frac{\dot{\theta}}{u^2} \qquad \dot{\theta} = u^2h$$

$$f = -h^2 u^2 \frac{du^2}{d\theta^2} - h^2 u^3 \quad (Eq\ 2)$$



a = Semi-major axis length

c = foci distance ; $c^2 = a^2 - b^2$

e = eccentricity $= \frac{c}{a}$

p = Semi-latis rectum

$$p = a(1-e^2) = \frac{b^2}{a}$$

Area $= \pi ab$

Equation of an ellipse in polar form:

$$r(\theta) = \frac{p}{1 + e\cos\theta}$$

$$u(\theta) = \frac{1 + e\cos\theta}{p} \qquad \frac{du}{d\theta} = \frac{-e}{p}\sin\theta \;,\; \frac{d^2u}{dp^2} = \frac{-e}{p}\cos\theta$$

$$f = -h^2u^2\frac{du^2}{d\theta^2} - h^2u^3 \rightarrow f = -h^2u^2\left[\frac{-e}{p}\cos\theta + \frac{1+e\cos\theta}{p}\right] \rightarrow f = \frac{-h^2u^2}{p} \rightarrow$$

$$f = \underbrace{\frac{-h^2}{p}}_{u} \cdot \frac{1}{r^2}$$ inverse Square of distance

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$$f = -\frac{h^2}{p} \cdot \frac{1}{r^2}$$

Kepler's 2nd law: $\frac{d\vec{A}}{dt} = \frac{1}{2}(\vec{r} \times \vec{v}) = \frac{\vec{h}}{2}$ Const

$$\frac{dA}{dt} = \frac{h}{2}$$

Area of ellipse: $A = \int_0^{T(\text{planet year})} \frac{h}{2}\, dt = \frac{hT}{2}$

$$h = \frac{2A}{T} \qquad A = \pi ab \qquad p = \frac{b^2}{a}$$

$$\frac{h^2}{p} = \left(\frac{2\pi ab}{T}\right)^2 \frac{a}{b^2} \qquad , \quad h = \frac{2\pi ab}{T}$$

$$\rightarrow \frac{h^2}{p} = \frac{4\pi^2 a^3}{T^2} \quad (\text{eq } 3)$$

Kepler's 3rd law: $T^2 \sim a^3$ where a is the mean radius of the ellipse

$\rightarrow \frac{a^3}{T^2}$ is the same for all the planets

$\frac{h^2}{p}$ is a property of Sun

$\frac{N}{kg} = \frac{kg\,m}{kg\,s^2}$ $\mu_\odot = \frac{h^2}{p}$ Sun's gravitational parameter

$f = -\frac{\mu_\odot}{r^2}$ gravitational attraction force per unit mass from the Sun

$\vec{f} = -\frac{\mu}{r^2}\hat{e}_r \qquad \vec{f} = -\frac{\mu \vec{r}}{r^3} \qquad \vec{F} = m\vec{f} = -\frac{\mu m}{r^2}\hat{e}_r$

$\ddot{\vec{r}} = -\frac{\mu \vec{r}}{r^3}$

Newton: $\mu \sim M$ mass of primary

$\mu = GM \quad , \mu_\oplus = GM_\oplus$; G is universal gravitational constant

Value of G found in 1798 by Cavendish

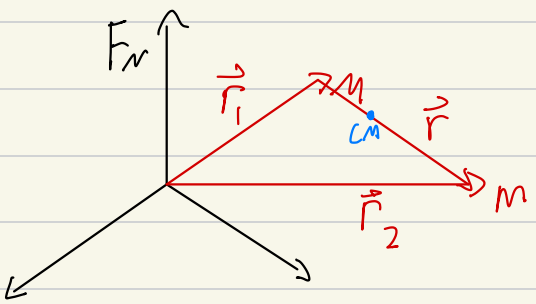
$\vec{F} = \frac{-G\overset{\mu}{M}m}{r^2}\hat{e}_r$ (Law of universal gravitation) $G = 6.674 \cdot 10^{-11} \frac{m^3}{kg \cdot s^2}$

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$F_{Gravity} = \frac{-GMm\hat{e}_r}{r^2}$

2-body problem

M mass of object #1, m mass of object #2



$\vec{F}_{gravity_2} = \frac{-GMm}{r^2} \cdot \frac{\vec{r}}{r}$

$\vec{F}_{gravity} = + \frac{GMm}{r^2} \cdot \frac{\vec{r}}{r}$

$$m\ddot{\vec{r}}_2 = \frac{-GMm}{r^2}\left(\frac{\vec{r}}{r}\right) \qquad M\ddot{\vec{r}}_1 = +\frac{GMm}{r^2}\left(\frac{\vec{r}}{r}\right)$$

Add the 2 eqns: $M\ddot{\vec{r}}_1 + m\ddot{\vec{r}}_2 = 0$

$M\dot{\vec{r}}_1 + m\dot{\vec{r}}_2 = \overrightarrow{\text{Constant}}$ → Constant vector

$$\vec{r}_{CM} = \frac{M\vec{r}_1 + m\vec{r}_2}{M+m}$$ (position of center of mass)

$$\dot{\vec{r}}_{CM} = \frac{M\dot{\vec{r}}_1 + m\dot{\vec{r}}_2}{M+m} = \overrightarrow{\text{Constant}}'$$ (different constant)

"Subtract" equations: (want to find $\ddot{\vec{r}} = \ddot{\vec{r}}_2 - \ddot{\vec{r}}_1$)

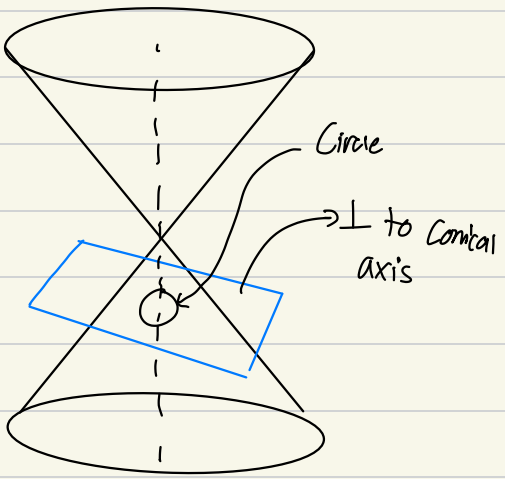
$$\ddot{\vec{r}} = \frac{-GM\vec{r}}{r^3} - \frac{Gm\vec{r}}{r^3} \rightarrow \ddot{\vec{r}} = \frac{-G(M+m)\vec{r}}{r^3}$$

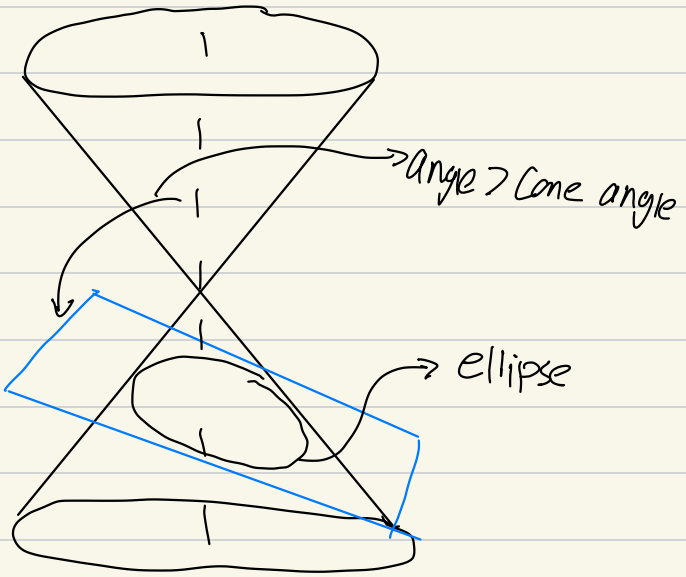
$$\ddot{\vec{r}} = -\mu\frac{\vec{r}}{r^2}$$

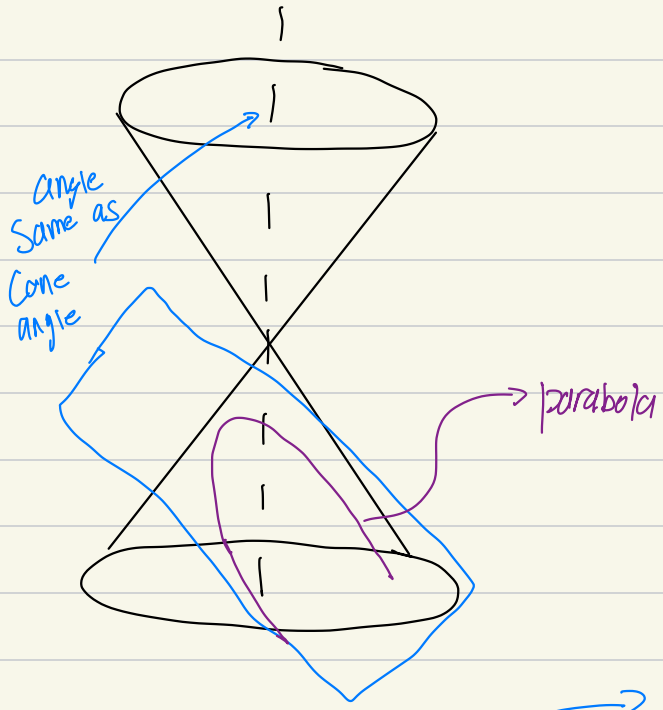
$\mu = G(M+m)$ for 2-body problem ; if $m \ll M$ then $\mu \approx GM$

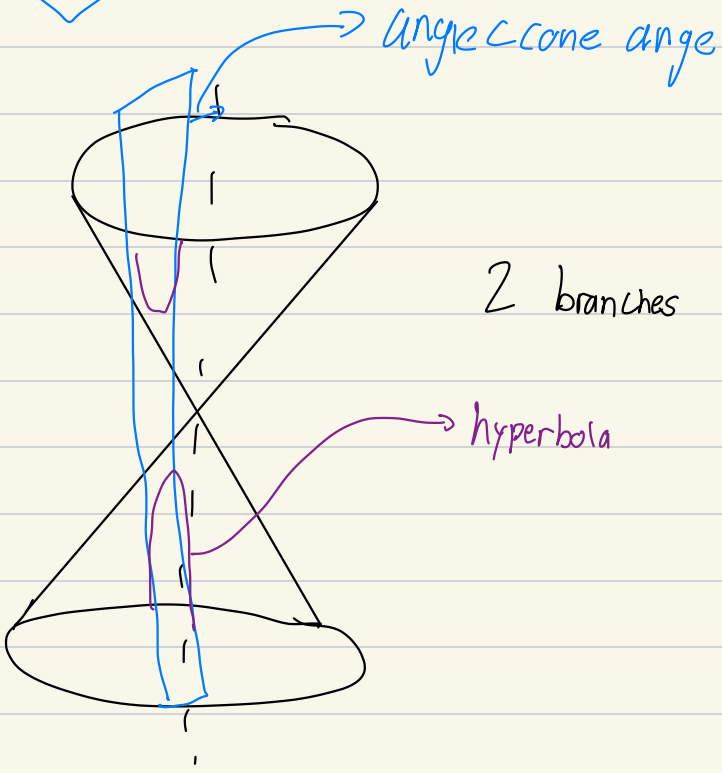
Kepler - Ellipse

Newton - Ellipse + all other conic sections for 2 body problem









e - eccentricity

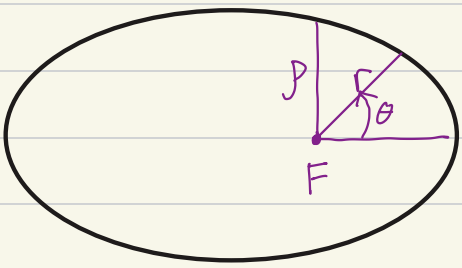
Circle $e = 0$

Ellipse $0 < e < 1$

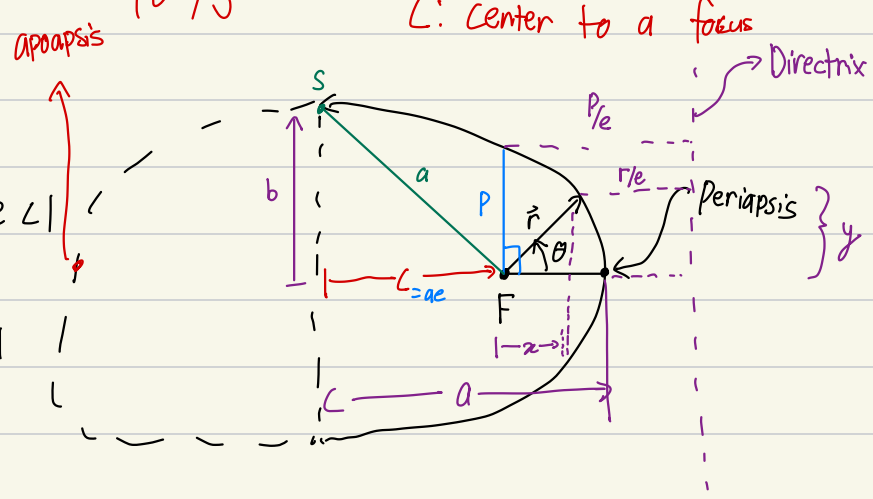
Parabola $e = 1$

Hyperbola $e > 1$

p Semi-latus rectum



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Circle $e=0$

Ellipse $0<e<1$

Parabola $e=1$

Hyperbola $e>1$

p = Semi-latus rectum

θ true anomaly

$$\frac{p}{e} = x + \frac{r}{e} \rightarrow p = xe + r \rightarrow p = r\cos\theta + r \rightarrow p = r(1+e\cos\theta) \rightarrow \boxed{r = \frac{p}{1+e\cos\theta}}$$

Minimum radius = periapsis radius $= r_p = \frac{p}{1+e}$ $(\theta = 0°)$

Maximum radius = apoapsis $= r_a = \frac{p}{1-e}$ $(\theta = \pi, 180°)$

Semi-major axis $2a = r_a + r_p = \frac{p}{1+e} + \frac{p}{1-e} = p\left(\frac{1-e+1+e}{1-e^2}\right) \rightarrow 2a = \frac{2p}{1-e^2} \rightarrow \boxed{p = a(1-e^2)}$

$a = \frac{r_a + r_p}{2}$ ("average radius")

$r_p = a(1-e)$

$r_a = a(1+e)$

Nomenclature	Body (F)	Periapsis	Apoapsis
Earth	⊕	Perigee	Apogee
Sun	⊙	perihelion	aphelion
Moon	☾	perilune / periselene	apolune / aposelene
Jupiter	21	Perijove	apojove

$C = a - r_p \rightarrow C = a - a(1-e) \rightarrow C = ae$

b = Semi-minor axis

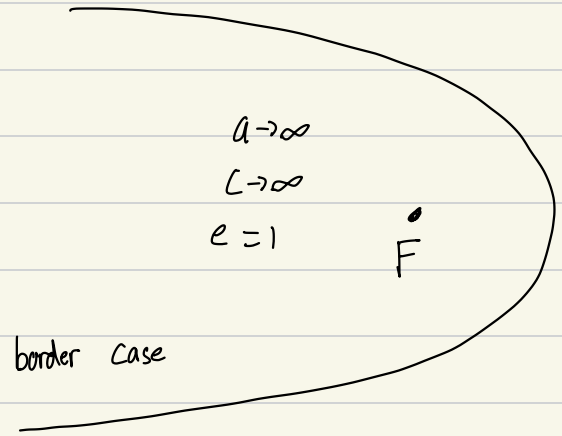
$$p = \frac{b^2}{a} \qquad \text{recall: } p = a(1-e^2)$$

$$b = \sqrt{pa}$$

$$b = a\sqrt{1-e^2}$$

$$|SF| = \sqrt{b^2 + c^2} \rightarrow \sqrt{a^2(1-e^2) + a^2e^2} \rightarrow |SF| = a$$

Parabolic trajectories

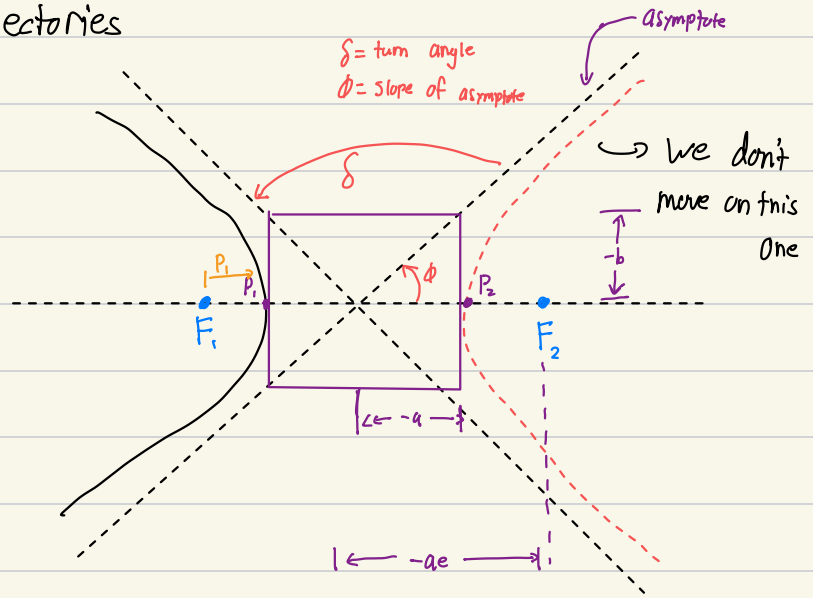


border case

btwn ellipse/hyperbola

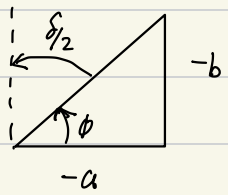
just barely doesn't return

Hyperbolic trajectories



δ - turn angle

ϕ - Slope of asymptote



$$\cos\phi = \frac{-a}{-ae} = \frac{1}{e}$$

$$\sin\left(\frac{\delta}{2}\right) = \frac{-a}{-ae} = \frac{1}{e}$$

$$\phi = \cos^{-1}\left(\frac{1}{e}\right)$$

$$b = a\sqrt{e^2 - 1}$$

Periapsis radius

$$r_p = -ae + a \rightarrow r_p = a(1-e)$$

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Constants of motion

- Angular momentum
- Specific Angular momentum

$\vec{h} = \vec{r} \times \vec{v}$

$\therefore$ orbit is in plane

$|h| = r^2 \dot{\theta}$ Kepler's 2nd law

Energy: K.E. per unit mass

$T = \frac{1}{2} v^2$; $v = |\vec{v}|$

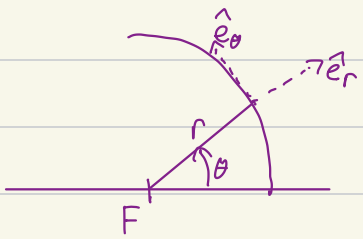
$= \frac{1}{2} \vec{v} \cdot \vec{v}$

$\dot{T} = \frac{1}{2}[\dot{\vec{r}} \cdot \dot{\vec{r}} + \dot{\vec{r}} \cdot \ddot{\vec{r}}]$ (product rule) $= \dot{\vec{r}} \cdot \ddot{\vec{r}}$

$\ddot{\vec{r}} = -\frac{\mu}{r^3} \vec{r}$

$= \dot{\vec{r}} \cdot \left(\frac{-\mu}{r^2} \hat{r}\right)$

$\dot{T} = \frac{-\mu}{r^3} (\dot{\vec{r}} \cdot \vec{r})$



$\vec{r} = r\hat{e}_r$

$\dot{\vec{r}} = \dot{r}\hat{e}_r + r\dot{\theta}\hat{e}_\theta$

$\dot{\vec{r}} \cdot \vec{r} = r\dot{r}$

$\dot{T} = \frac{-\mu}{r^3} r\dot{r} = -\frac{\mu}{r^2} \dot{r}$

$$\frac{d}{dt}(T) = \frac{d}{dt}\left(\frac{\mu}{r}\right)$$

$$\frac{d}{dt}\left(T - \frac{\mu}{r}\right) = 0 \rightarrow \underbrace{\frac{v^2}{2}}_{\text{K.E. per unit mass}} - \overbrace{\frac{\mu}{r}}^{\text{P.E. per unit mass}} = \text{Constant}$$

$T \qquad V$

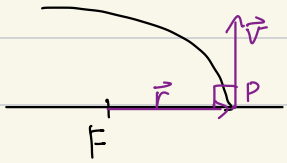
Choose $V = -\frac{\mu}{r}$ to be the gravitational energy (per unit mass) for an orbit

$$E = \frac{v^2}{2} - \frac{\mu}{r}$$

$V = 0$ as $r \rightarrow \infty$ & V is less than zero everywhere else

Relate this to conic section parameters

$$\vec{h} = \vec{r} \times \vec{v} \; ; \; |h| = r_p v_p \quad \text{(based on periapsis)}$$



note: $p = \frac{h^2}{\mu}$ (will be proven later)

$$p = a(1-e^2) \rightarrow h = \sqrt{\mu p} \rightarrow h = \sqrt{\mu a(1-e^2)}$$

$$\overbrace{\sqrt{\mu a(1-e^2)}}^{h} = \overbrace{a(1-e)}^{r_p} v_p \rightarrow v_p^2 = \frac{\mu a(1-e^2)}{a^2(1-e^2)}$$

$$E = E_p = \frac{v_p^2}{2} - \frac{\mu}{r_p} = \frac{\mu(1-e^2)}{a(1-e)^2} - \frac{\mu}{a(1-e)} \rightarrow E = -\frac{\mu}{2a}$$

$$E = \frac{v^2}{2} - \frac{\mu}{r} = -\frac{\mu}{2a}$$

$E < 0$ ellipse

$E = 0$ parabola

$E > 0$ hyperbola

For a circle: $r = a$

$$\frac{v^2}{2} - \frac{\mu}{r} = \frac{-\mu}{2r} \longrightarrow v^2 = \frac{\mu}{r}$$

End of content for midterm 1

Eccentricity vector:

Consider $\vec{r} \times \vec{h}$

$$\frac{d}{dt}(\vec{r} \times \vec{h}) = (\dot{\vec{r}} \times \vec{h}) + (\vec{r} \times \dot{\vec{h}})$$

$\dot{\vec{h}} = 0$ since $\vec{h} = \text{const.}$

$$= \frac{-\mu}{r^3} \vec{r} \times (\vec{r} \times \dot{\vec{r}})$$

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Eccentricity vector:

$$\frac{d}{dt}\left(\dot{\vec{r}} \times \vec{h}\right) = -\frac{\mu}{r^3}\vec{r} \times (\vec{r} \times \dot{\vec{r}})$$

vector identity: $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$

$$\frac{d}{dt}(\dot{\vec{r}} \times \vec{h}) = -\frac{\mu}{r^3}\left[(\vec{r} \cdot \dot{\vec{r}})\vec{r} - (\vec{r} \cdot \vec{r})\dot{\vec{r}}\right]$$

where $\vec{r} = r\hat{e}_r$

$$\dot{\vec{r}} = \dot{r}\hat{e}_r + r\dot{\theta}\hat{e}_\theta$$

$$\vec{r} \cdot \dot{\vec{r}} = r\dot{r}$$

$$= \frac{\mu}{r^3}\left[(r\dot{r})\vec{r} - r^2\dot{\vec{r}}\right] = \frac{\mu}{r^2}(\dot{r}\vec{r} - r\dot{\vec{r}})$$

note: $\frac{d}{dt}\left(\frac{\vec{r}}{r}\right) = \frac{r\dot{\vec{r}} - \vec{r}\dot{r}}{r^2}$

$$\frac{d}{dt}(\dot{\vec{r}} \times \vec{h}) = \mu\frac{d}{dt}\left(\frac{\vec{r}}{r}\right)$$

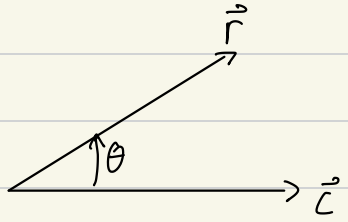
$$\frac{d}{dt}\left(\dot{\vec{r}} \times \vec{h} - \mu\frac{\vec{r}}{r}\right) = 0 \quad \therefore \quad \dot{\vec{r}} \times \vec{h} - \mu\frac{\vec{r}}{r} = 0$$

$$\dot{\vec{r}} \times \vec{h} - \mu \frac{\vec{r}}{r} = \vec{C} \qquad \text{(constant vector)}$$

$$\vec{r} \cdot \vec{c} = \vec{r} \cdot \left(\dot{\vec{r}} \times \vec{h} - \mu \frac{\vec{r}}{r} \right) = r\hat{e}_r \cdot \left[(\dot{r}\hat{e}_r + r\dot{\theta}\hat{e}_\theta) \times h\hat{e}_h - \mu \hat{e}_r \right] =$$

$$= r\hat{e}_r \cdot \left[-\dot{r}h\hat{e}_\theta + r\dot{\theta}h\hat{e}_r - \mu \hat{e}_r \right] = \underbrace{r^2\dot{\theta}}_{h} h - \mu r$$

$$= h^2 - \mu r$$



$c = |\vec{c}|$

$$\vec{r} \cdot \vec{c} = rC\cos(\theta) = h^2 - \mu r$$

$$r = \frac{\frac{h^2}{\mu}}{1 + \frac{C}{\mu}\cos\theta}$$

$$r = \frac{p}{1 + e\cos\theta}$$

$$p = \frac{h^2}{\mu} \qquad e = \frac{C}{\mu}$$

$$\vec{C} = \dot{\vec{r}} \times \vec{h} - \mu \frac{\vec{r}}{r}$$

$$\frac{\vec{C}}{\mu} = \frac{\vec{v} \times \vec{h}}{\mu} - \frac{\vec{r}}{r}$$

eccentricity vector: $\vec{e} = \frac{\vec{v} \times \vec{h}}{\mu} - \frac{\vec{r}}{r}$ Points to periapsis $e = |\vec{e}|$

$$E = \frac{v^2}{2} - \frac{\mu}{r} = \frac{-\mu}{2a}$$

for circle $a = r =$ Constant

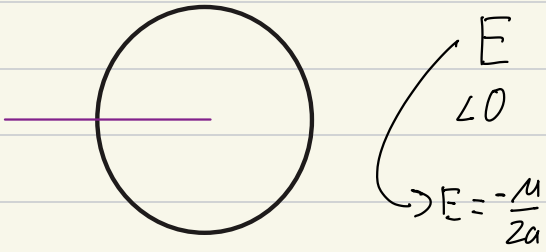
$$v_{circle}^2 = \frac{\mu}{r}$$

for escape/parabola, $E = 0$

$$E = 0 = \frac{v_{esc}^2}{2} - \frac{\mu}{r} \rightarrow v_{esc}^2 = \frac{2\mu}{r}$$

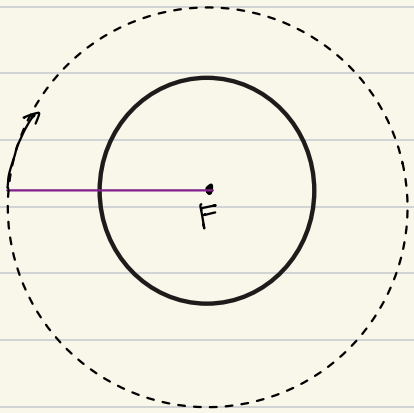
- independent of direction
- but don't hit the planet!!

$$v_{esc} = \sqrt{2}\, v_{circle}$$

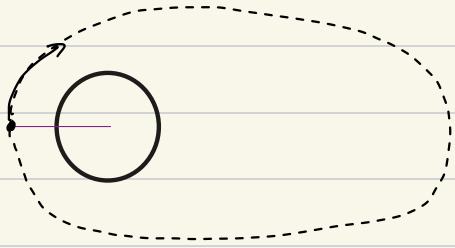


E	e	a
<0	$0<e<1$	>0

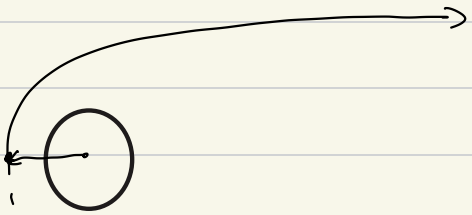
$E = -\frac{\mu}{2a}$



E	e	a
<0	0	>0



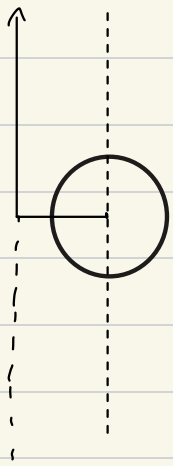
E	e	a
<0	$0<e<1$	>0



E	e	a
$=0$	$=1$	$\to\infty$



E	e	a
>0	>1	<0



E	e	a
$\to\infty$	$\to\infty$	$\to 0^-$

Orbital frame $\{F, \hat{e}_r, \hat{e}_\theta, \hat{e}_n\}$

Perifocal frame $\{F, \hat{p}, \hat{q}, \hat{p}\times\hat{q}\}$

